AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS X EXAMINATION 2010

Computer Science Paper II

Time allowed: 2 hours 25 minutes Marks 40

INSTRUCTIONS

Please read the following instructions carefully.

1. Check your name and school information. Sign that it is correct.

I agree that this is my name and school. **Candidate's signature**

2. RUBRIC. There are EIGHT questions. Answer ALL EIGHT questions. Questions 7 & 8 each offer TWO choices. Attempt any ONE choice from each.

3. When answering the questions:

   Read each question carefully.
   Use a black pencil for diagrams. DO NOT use coloured pencils.
   DO NOT use staples, paper clips, glue, correcting fluid or ink erasers.
   Complete your answer in the allocated space only. DO NOT write outside the answer box.

4. The marks for the questions are shown in brackets ( ).

Q.1. (Total 5 Marks)

When there is a requirement to replace a manual system by computers, then a problem has to be studied and identified as one that could be solved by using a computer. Some stages of a computer solution are listed below but not in the right order.

a. Put the appropriate number beside each step to represent the right order: (2 Marks)

Solution planning _____
Problem definition _____
Implementation _____
Solution design _____
Testing _____

b. Imagine that you are a system analyst. Prepare a solution to the problem related to the use of computers for student's registration at your school. Identify at least THREE stages. (3 Marks)

Stage 1_____

_____

Stage 2_____

_____

Stage 3_____

_____

Q.2. (Total 5 Marks)

Assign an appropriate variable that may be used in BASIC language beside each data given in the table below:

Field	Data	Variable
Name	Farida	
Telephone	0921-35224766	
Roll Number	0910201	
Date of birth	10/06/19195	
Registration year	2008	

Q.3. (Total 5 Marks)

Consider the given program

```
10  ClS
20  FOR I= 1 to 5
30  J=J+ I
40  _____________
50  NEXT I
```

a. What will be the output if you insert the given statements at line number 40? (3 Marks)

i. PRINT I _____

ii. PRINT J;_____

iii. PRINT J*I, _____

b. What will be the output if STEP 2 is added in line 20? (2 Marks)

i. PRINT I_____

ii. PRINT J*I_____

Q.4. (Total 4 Marks)

Write a program to read the following numerical information and print the given array in the same format.

5	6	5	3
7	4	7	1
7	6	5	3
6	4	8	2

Q.5. (Total 3 Marks)

Define the function for finding the area of a circle.

Q.6. (Total 3 Marks)

Construct a truth table for the following Boolean expression:

$F = A\,\overline{B} + \overline{A}\,C$

Row	A	B	C	F

Q.7. (Total 8 Marks)

EITHER

a. Discuss the ways in which computers have made cricket more interesting to watch on television.

OR

b. What are some ways in which computers are helpful to scientists?

Q.8. (Total 7 Marks)

EITHER

a. Write a program to prepare a marks sheet for candidates. Mention personal information, marks obtained out of 100 for five subjects. Find the total and display the grade beside the total.

Keep the grade criteria as following.

80-100	A
60-79	B
40 -59	C

OR

b. Write a program to calculate the area of various geometrical shapes; a rectangle, a triangle and a circle. The program must use multiple branch statement and input must be received from the keyboard.

END OF PAPER